INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

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THE SITE

Location	T5 Chicago II, IL -- station KORD, America/Chicago
Committed pair	OSM 1219083554 -> 1219083552 T5 Chicago II / OSM way 1219083552
Facade gap	73.4 m between the two halls
Weather record	43,775 real hourly records from KORD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.7306 C at 260 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-07-31 (knife_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 2 mode changes. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0.

Agent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	5 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

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hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	19.589	21.0	18.900	1.326
01	FREE	yes	-	19.309	21.0	18.340	1.278
02	FREE	yes	-	19.309	21.0	18.340	1.240
03	FREE	yes	-	18.754	21.0	17.230	1.176
04	FREE	yes	-	18.204	21.0	16.680	1.124
05	FREE	yes	-	18.204	21.0	16.680	1.029
06	FREE	yes	-	17.929	21.0	16.680	1.075
07	mechanical	no	dew point	19.933	21.0	19.978	1.438
08	mechanical	no	dry-bulb	21.525	21.0	20.001	2.079
09	mechanical	no	refusal	23.105	21.0	22.540	2.032
10	mechanical	no	refusal	25.155	21.0	23.435	1.958
11	mechanical	no	dry-bulb	24.869	21.0	23.340	1.645
12	mechanical	no	refusal	25.981	21.0	24.599	1.630
13	mechanical	no	refusal	26.285	21.0	25.709	1.419
14	mechanical	no	refusal	26.159	21.0	26.215	1.160

15	mechanical	no	refusal	26.390	21.0	26.819	1.200
16	mechanical	no	refusal	26.850	21.0	26.819	1.366
17	mechanical	no	refusal	27.138	21.0	26.221	1.180
18	mechanical	no	refusal	26.203	21.0	26.057	1.220
19	mechanical	no	refusal	25.949	21.0	24.937	1.471
20	mechanical	no	dry-bulb	24.978	21.0	24.039	1.353
21	mechanical	no	refusal	24.534	21.0	23.277	1.325
22	mechanical	no	refusal	24.230	21.0	22.929	1.180
23	mechanical	no	refusal	23.258	21.0	21.890	1.300

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.589 C, which is 1.411 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.326 C, and the margin is measured, not chosen: 0.912 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.309 C, which is 1.691 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.278 C, and the margin is measured, not chosen: 0.863 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.309 C, which is 1.691 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.240 C, and the margin is measured, not chosen: 0.825 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.754 C, which is 2.246 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.176 C, and the margin is measured, not chosen: 0.762 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.204 C, which is 2.796 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.124 C, and the margin is measured, not chosen: 0.710 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.204 C, which is 2.796 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.029 C, and the margin is measured, not chosen: 0.615 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.929 C, which is 3.071 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.075 C, and the margin is measured, not chosen: 0.661 C of group-conditional forecast error for this hour of day, plus 0.4144 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 19.933 C would have passed. The outside air is too HUMID: the dew-point bound is 15.03 C against a 15.0 C maximum, failing by 0.03 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.525 C against a 21.0 C limit, so it fails by 0.525 C. It would take a limit 0.525 C higher, or a bound 0.525 C tighter, to change this hour.

09:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

10:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.869 C against a 21.0 C limit, so it fails by 3.869 C. It would take a limit 3.869 C higher, or a bound 3.869 C tighter, to change this hour.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

19:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.978 C against a 21.0 C limit, so it fails by 3.978 C. It would take a limit 3.978 C higher, or a bound 3.978 C tighter, to change this hour.

21:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

22:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

23:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

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